# MARKERLESS HUMAN MOTION ANALYSIS UNDER THERMAL STRESS USING OPTICAL DEPTH AND MMWAVE RADAR

## The Craton Suite

The Craton Suite is a markerless motion tracking system designed to analyze human kinematics under extreme thermal stress. Traditional optical motion capture systems degrade under the irregular movements and environmental constraints introduced by heat-stressed exercise. To address this, the system provides a non-invasive alternative to laboratory-bound motion capture.

## HOW IT WORKS?

### 1. Dual-Sensor Acquisition

Integrates an Intel RealSense D435i depth camera with a Texas Instruments IWR6843L mmWave radar to capture complementary motion data.

### 2. Data Processing

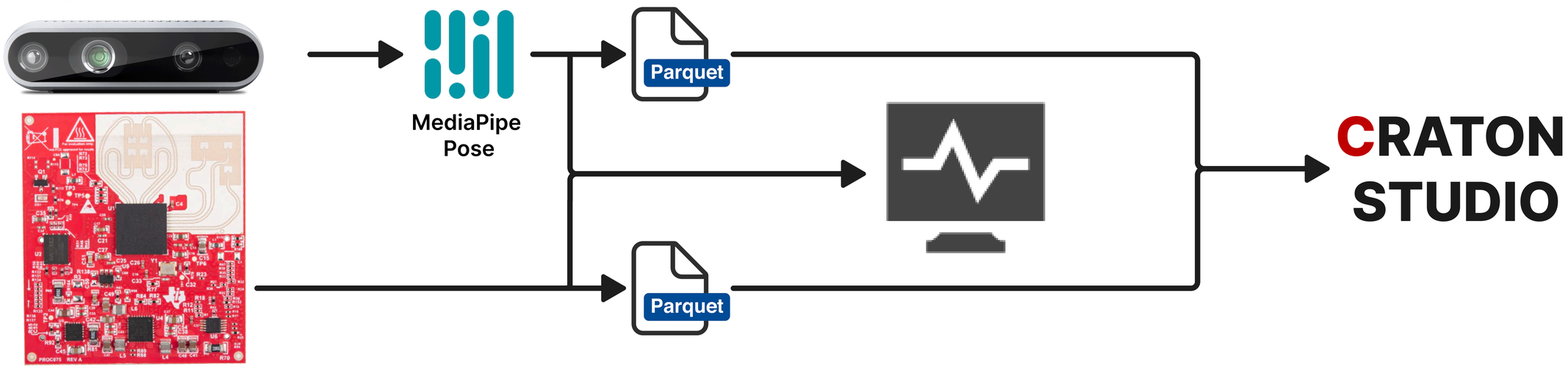
Camera: Utilizes MediaPipe to extract 3D joint coordinates
Radar: Logs RDHM data for post-session analysis.

### 3. Monitoring

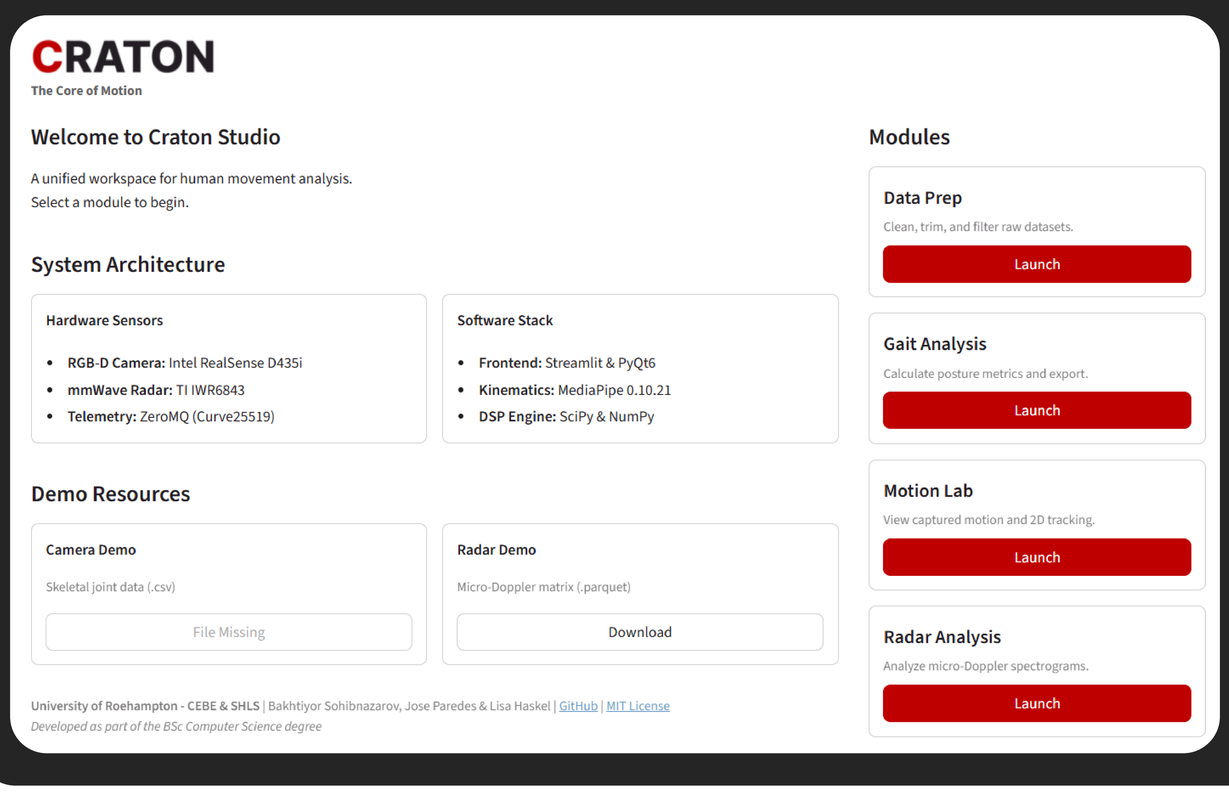
Processed optical kinematics and raw radar telemetry are securely transmitted over LAN for remote session monitoring

### 4. Metrics Calculation

Data is fused offline to calculate specific bio-mechanical markers, such as variations in trunk lean and micro-Doppler gait cadence.




## Craton Studio: The Clinical Dashboard




### Edge-Deployed Dashboard

A custom-built Streamlit application designed to run locally on standard commercial laptops, eliminating the need for complex web overhead or heavy GPU clusters.

### Offline Late-Fusion Processing

It processes previously acquired datasets offline, applying signal filtering to independent optical and radar streams to calculate specific bio-mechanical markers like posture stability and cadence.

### Data Visualization and Export

It visualizes the temporally aligned metrics side-by-side for clinical comparison and exports the final calculated datasets as formatted CSV files for further evaluation.

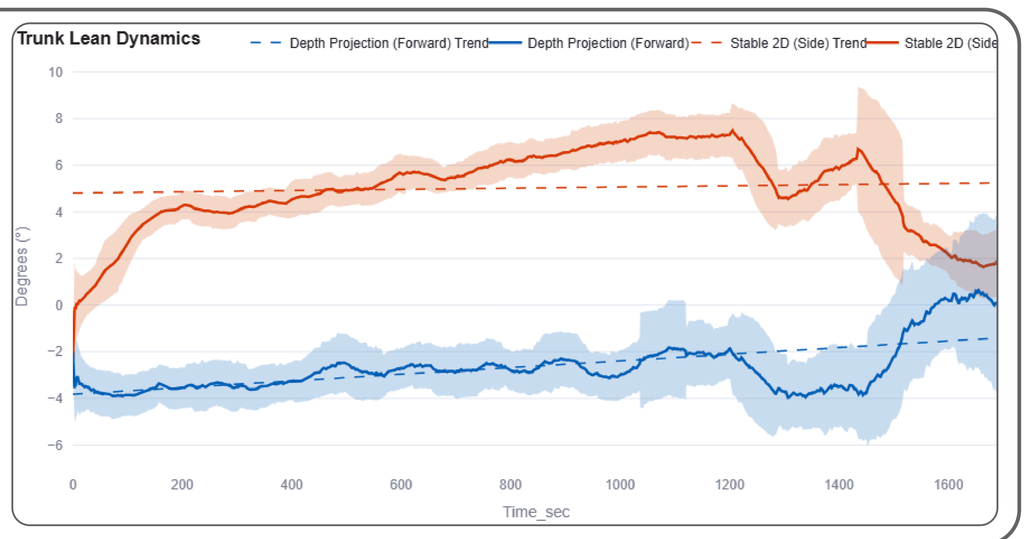
## Demo Analysis: Trunk Lean Dynamics

### Visualizing Fatigue

Tracks real-time postural deviations over a 1600-second physical exertion period.

### Insight

Shaded variance bands clearly highlight instances of mechanical breakdown and instability as thermal stress increases.





**REFERENCES**

[1] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," arXiv preprint arXiv:1906.08172, 2019.